

Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Structures for Energy Absorption

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Tensegrity-inspired architectures combine rigid compression members with a continuous network of soft tension elements, yielding lightweight assemblies with tunable nonlinear responses that are attractive for energy-absorbing applications such as protective equipment, packaging, and aerospace landing structures. Multi-material fused-deposition modeling (FDM) of polylactic acid (PLA) struts and thermoplastic polyurethane (TPU) tension elements makes it possible to fabricate diverse tensegrity-inspired geometries rapidly enough to support a high-throughput experimental search, while the rate-dependent damping of TPU is itself an asset for impact mitigation. We report an experiment-driven Bayesian optimization (BO) framework in which a Gaussian-process surrogate is trained directly on physical quasi-static compression and drop-weight impact data and used to recommend the next batch of designs to print and test. The framework parameterizes a family of tensegrity-inspired unit cells over strut geometry, tension-element cross-section, connectivity topology, and unit-cell tiling, and optimizes peak transmitted force, specific energy absorption, and compaction efficiency.

Tighten abstract to 150–200 words with concrete numerical results once the experimental campaign is complete.

Keywords: tensegrity, multi-material 3D printing, Bayesian optimization, energy absorption, design of experiments

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1 Introduction

Manuscript status. At time of writing the project is at the *planned-methods* phase: the parameterized unit cell, multi-material print recipe, joint geometry, quasi-static and drop-impact protocols, and Bayesian-optimization loop described in Section 3 have been scaffolded and prototype hardware has been printed, but the closed-loop experimental campaign has not yet produced reportable results. Sections 4 and 5 therefore remain placeholders.

Tensegrity structures are assemblies of rigid compression members suspended within a continuous network of tension members; their defining property—no two rigid bars touch—yields lightweight assemblies with remarkable strength-to-weight ratios and tunable nonlinear mechanical responses [1, 2]. These properties have made tensegrity attractive across scales, from meta-material unit cells [3–5] to compliant impact-absorbing robots developed for planetary landing [6–10]. Recent work has shown that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact [11], opening a path toward additively manufactured architected absorbers.

Multi-material fused-deposition modeling (FDM) using polylactic acid (PLA) for rigid struts and thermoplastic polyurethane (TPU) for flexible tension elements enables rapid fabrication of diverse tensegrity-inspired geometries on a single platform [12, 13]. Although extruded TPU does not behave as an idealized inextensible cable, its rate-dependent damping is a desirable property for impact-absorption applications and complements the elastic stiffness of the PLA compression members rather than competing with it.

Designing such architected absorbers is fundamentally a *design-of-experiments* problem: the configuration space spanned by strut geometry, tension-element cross-section, connectivity, and unit-cell tiling is large; each candidate must be physically printed

and tested; and the relevant performance metrics—peak transmitted force, specific energy absorption (SEA), and compaction efficiency—are noisy and partially correlated. Bayesian optimization (BO) is well-matched to this regime, having driven breakthroughs from hyperparameter tuning of AlphaGo [14] to sustainable concrete [15], autonomous discovery of organic laser emitters [16], and architected materials [17–20].

Contributions. This paper makes the following contributions:

- (1) A parameterized family of multi-material 3D-printable tensegrity-inspired unit cells with PLA compression members and TPU tension elements wrapped using the core-encapsulation strategy of Ye et al. [12] to ensure cyclic interface durability.
- (2) An experiment-driven BO loop, built on BoTorch/Ax [21], that operates directly on physical quasi-static compression and drop-weight impact measurements, recommending the next designs to fabricate without requiring an accurate forward simulator.
- (3) A two-fidelity escalation path that pairs the direct-experimental loop above with a planned higher-fidelity rung based on *pretensioned* tensegrity assemblies (true cables and measured pretension) for selected Pareto-optimal designs; the specific surrogate / fusion strategy is left open.

Figure overview: Closed-loop, experiment-driven design framework: parameterized PLA/TPU tensegrity-inspired unit cells are printed, tested under quasi-static compression and drop-weight impact, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs.

The remainder of the paper is organized as follows. Section 2 reviews tensegrity mechanics, multi-material 3D printing for energy absorption, and Bayesian optimization with an emphasis on multi-objective and noise-robust acquisition functions. Section 3

Add 1–2 sentences clearly stating the gap this paper closes vs. Pajunen et al. [11] (single fabrication condition, no closed-loop optimization) and Mo et al. [17] (multi-fidelity BO, but simulation-only)

Decide between (a) co-Kriging or nonlinear information-fusion priors [22] once pretensioned-assembly data are available, or (b) keep this work single-fidelity and report the pretensioned campaign separately.

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Fig. 1 Placeholder. Closed-loop, experiment-driven design framework: parameterized PLA/TPU tensegrity-inspired unit cells are printed, tested under quasi-static compression and drop-weight impact, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs.

introduces the design parameterization, fabrication workflow, experimental protocols, and BO loop. Section 4 reports the results of the closed-loop experimental campaign, and Section 5 discusses the implications and limitations of the approach. Section 6 concludes.

2 Background

2.1 Tensegrity-Inspired Architectures for Energy Absorption. Tensegrity prisms and their planar extensions exhibit softening–stiffening response under compression [3–5], characteristics desirable for impact mitigation because the structure can absorb a large fraction of the input energy at a controlled, sub-injurious force plateau. Santos [23] recently characterized a tensegrity energy-dissipation metamaterial reporting equivalent viscous damping on the order of ~23% with negligible residual strain per impact, underscoring the suitability of tensegrity-inspired architectures for repeated-impact use cases. The space program literature documents large-scale tensegrity-inspired absorbers for planetary landing [6–10] and a long history of crushable energy-absorbing landing systems [24–26]. Pajunen et al. [11] demonstrated that truncated-tetrahedral tensegrity-inspired unit cells can be directly 3D-printed and exhibit load-limiting plateaus, motivating the present focus on additively manufactured TPU/PLA composites. More recent work by Davami et al. on dynamically loaded AM tensegrity unit cells, with Intrigila et al. as the closest published analog (SLA Tough 2000, double-T3 quasi-static), confirms that the same architecture class also responds favorably under drop-impact loading, although neither study employed multi-material FFF or closed-loop design optimization.

2.2 Multi-Material 3D Printing of PLA/TPU Composites. Multi-material rigid–soft FDM [12, 13] enables PLA–TPU combinations on a single platform with tunable energy absorption. Crucially, Ye et al. [12] introduced a *wrapping-based* strategy in which rigid cores are encapsulated by continuous soft skins, preventing interface delamination and enabling cyclic durability—a critical enabler for our fabrication approach. Recent multi-material PLA/TPU sandwich and layered composites [27, 28] report quantitative bounds on stiffness, strength, and energy absorption for the same PLA–TPU pair used here, while FFF-PLA fatigue data [29, 30] inform conservative design bounds for the rigid struts. Architected multi-material lattices with co-printed compliant skins have been explicitly designed for high specific energy absorption [31], with FFF-specific multimaterial interface adhesion [32]; recent surveys catalogue the rapidly expanding additively manufactured polymeric energy-absorption literature [33, 34].

2.3 Bayesian Optimization for Architected-Material Design. Bayesian optimization is a sequential, model-based strategy

for optimizing expensive black-box functions using a Gaussian-process surrogate [35, 36]. Modern implementations such as BoTorch [21] support parallel batched evaluations and a wide variety of acquisition functions; recent algorithmic advances include log-EI for numerical robustness [37], noisy expected hypervolume improvement for multi-objective settings [38], input-noise-robust acquisitions [39], and evolution-guided constrained multi-objective BO for self-driving labs [40]. Mixed/categorical search spaces—needed here for the discrete connectivity-topology variable—have dedicated treatments in BOCS [41] and the one-hot-with-rounded-kernel construction of [42]. BO has been applied to mechanical metamaterial and structural design [17–19] and to additive manufacturing [20]. Mo et al. [17] in particular established a multifidelity BO framework for architected materials by fusing low-fidelity simulations with high-fidelity ground truth via non-linear information-fusion priors [22]; the present work shifts the emphasis from simulation surrogates to direct experimental data, appropriate for materials whose constitutive response is difficult to calibrate from first principles.

3 Materials and Methods

3.1 Design Parameterization. We parameterize a family of tensegrity-inspired unit cells via four groups of design variables:

- **PLA compression members.** Strut diameter d_s and length ℓ_s , with the slenderness ratio ℓ_s/d_s bounded away from buckling-dominated regimes.
- **TPU tension elements.** Cross-sectional area A_t (or equivalent diameter d_t for circular cross-sections) and pre-tension geometry imposed during printing.
- **Connectivity topology.** Selected from a discrete set of candidate unit cells (e.g., truncated-tetrahedral, prism-based, and octahedral variants); the design vector encodes a categorical choice.
- **Unit-cell tiling.** Number of cells $N_x \times N_y \times N_z$ and the orientation of each cell relative to the impact direction.

Working prototype. The first instance of this family used to validate the fabrication and test pipeline is a three-strut (T_3) tensegrity prism scaled to a ~50 mm bounding box, with $d_s =$ and a cable diameter d_t swept over the FFF-resolvable set {1.2, 1.8, 2.4, 3.0, 4.5} mm. Under the prism’s D_3 symmetry the twelve member-diameter axes collapse to four orbit axes (one strut orbit and three cable orbits: saddle, top, bottom), which keeps the initial design vector low-dimensional while leaving room to relax to per-member diameters once the saddle/top/bottom orbits have been individually characterized.

Table 1 Placeholder. Design variables, units, lower/upper bounds, manufacturing feasibility constraints, and categorical encoding.

Column A	Column B	Column C
(table contents pending)		

Table design-vars: Design variables, units, lower/upper bounds, manufacturing feasibility constraints, and categorical encoding.

3.2 Multi-Material Fabrication. Specimens are fabricated on a multi-material FDM printer using PLA for the rigid struts and TPU for the soft tension elements. Following Ye et al. [12], the design uses a core-wrapping strategy in which each PLA strut is encapsulated by continuous TPU skin to provide robust mechanical interlocking and prevent interface delamination during cyclic loading.

Print platform and joint geometry... The working platform is a Bambu Lab H2D IDEX printer, which permits a single rigid–soft build without filament swaps. Strut endpoints are tied to cables through parametric joint geometries developed and ranked through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants); the working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies, with a captive TPU tendon inside the PLA shell to keep the soft tendon protected from line failure at the strut-to-cable transition.

Supports for soft members... Because near-vertical members are otherwise unsupported during printing, the slicer profile combines a tensegrity-specific Bambu Studio recipe (support threshold angle dropped from 40° to 10°, support material matched to the rigid extruder) with manually generated narrowing pillars that ray-cast against the printable mesh underside, producing TPU-safe coverage of vertical members without slicer-side painting.

Figure placeholder: Fabrication workflow: parameter → CAD → slicer → multi-material print → post-process and inspect.

Fig. 2 Placeholder. Fabrication workflow: parameter → CAD → slicer → multi-material print → post-process and inspect.

Figure fab-workFlow: Fabrication workflow: parameter → CAD → slicer → multi-material print → post-process and inspect.

3.3 Experimental Characterization. Each printed specimen is subjected to two physical tests:

Quasi-static compression... A uniaxial load–displacement curve is recorded on a benchtop electromechanical load frame (BYU CB 123 Instron) following best-practice combinations of ASTM D638, ASTM D412, ASTM E111, ASTM F2971, and ISO/ASTM 52900/52921 adapted to architected polymer cells. Because expected cell stiffnesses fall in the ~5–30 g force range, the default 5 kN load cell is over-ranged and is being replaced by a 100–500 N cell for the first campaign. Reported metrics are

$$SEA = \frac{1}{m} \int_0^{\delta_d} F(\delta) d\delta, \quad \eta_c = \frac{\int_0^{\delta_d} F(\delta) d\delta}{F_{\max} \delta_d}, \quad (1)$$

where m is the specimen mass, δ_d is the densification displacement, $F_{\max}$ is the peak transmitted force, and η_c is the compaction efficiency.

Drop-weight impact... Following Pajunen et al. [11], specimens are subjected to drop-weight impact tests with instrumented load cells; we report the peak transmitted force and the energy-absorption plateau characteristics. The primary fixture is a bungee-assisted laboratory drop tower in which the base accelerates downward faster than 1g; because unconstrained specimens lift off the base during descent by design of the rig, the protocol constrains the specimen top through light tethers (capping upward travel) and registers the base via transfer tape or V-block features so that loading-direction compliance is not altered. A higher-fidelity

replicate campaign on a Lansmont M23 drop tower with synchronized Polytec QTEC laser vibrometry is planned for selected Pareto-optimal designs. Slip resistance and traction at the specimen–floor interface—governed by the tip geometry rather than the internal lattice—follow the test methodology of ISO 11334-4 [43] for walking-aid tips and are out of scope for the present axial-impact study.

Per-modality objectives... Five complementary data streams (triaxial accelerometer, high-speed video, shaker transfer function, slug-firing gas gun, and Polytec QTEC laser-Doppler vibrometer) each contribute a distinct raw observable; the mapping from each stream to BO-ready scalar objectives (SEA, VEA, $F_{\max}$, cushion-curve intercept, Gibson–Ashby relative density) is defined in five companion briefs and consolidated in a cross-modality synthesis.

3.4 Experiment-Driven Bayesian Optimization Loop. The optimization loop maintains a Gaussian-process (GP) surrogate over the design space defined in Section 3.1. After each round of physical experiments, the GP is updated and an acquisition function is maximized to recommend the next batch of designs to fabricate. Implementation uses BoTorch [21] with the log-Expected-Improvement (LogEI) acquisition [37] for the single-objective baseline and noisy Expected Hypervolume Improvement (qNEHVI) [38] for the multi-objective case (simultaneously maximizing SEA and compaction efficiency while constraining peak transmitted force). When experimental noise dominates (e.g., between-print variability), we additionally evaluate the input-noise-robust acquisition of Daulton et al. [39] and the evolution-guided constrained variant of Low et al. [40]. The discrete connectivity-topology variable is handled with the mixed-search-space construction of [42]; the BOCS combinatorial baseline of [41] is reported for comparison.

Implementation status... The campaign harness is built on Ax’s HierarchicalSearchSpace so that printer-feasibility constraints (e.g. the cable-diameter categorical depending on the chosen connectivity topology) can be encoded directly in the search space rather than enforced post hoc. The first physical batch consists of $n = 9$ specimens generated by a Sobol sequence over the four D_3 -orbit axes of the working T_3 prism and printed in a single 3×3 plate layout. For the present ≤ 25 -dimensional parameterization the default acquisition pairing is SAASBO with qNEHVI; escalation to TuRBO is planned only if larger per-member-diameter tilings push the design space well beyond this regime.

Figure placeholder: Closed-loop BO schematic: design proposal → multi-material print → compression and drop-weight tests → GP update and acquisition maximization → next proposal.

Fig. 3 Placeholder. Closed-loop BO schematic: design proposal → multi-material print → compression and drop-weight tests → GP update and acquisition maximization → next proposal.

Figure bo-loop: Closed-loop BO schematic: design proposal → multi-material print → compression and drop-weight tests → GP update and acquisition maximization → next proposal.

4 Results

Populate Results section once the experimental campaign produces data. Suggested subsections follow.

4.1 Convergence of the BO Loop.

Figure placeholder: Best-so-far SEA (and Pareto hypervolume for the multi-objective runs) vs. number of physical experiments, averaged over random seeds; baselines: LHS and random search.

Fig. 4 Placeholder. Best-so-far SEA (and Pareto hypervolume for the multi-objective runs) vs. number of physical experiments, averaged over random seeds; baselines: LHS and random search.

Figure convergence: Best-so-far SEA (and Pareto hypervolume for the multi-objective runs) vs. number of physical experiments, averaged over random seeds; baselines: LHS and random search.

4.2 Pareto-Optimal Designs.

Figure placeholder: Pareto front in (peak transmitted force, SEA) space; representative geometries annotated.

Fig. 5 Placeholder. Pareto front in (peak transmitted force, SEA) space; representative geometries annotated.

Figure pareto: Pareto front in (peak transmitted force, SEA) space; representative geometries annotated.

4.3 Reproducibility Across Print Replicates.

Table 2 Placeholder. Mean and CV of SEA, peak force, and compaction efficiency across n replicate prints for the top- k designs.

Column A	Column B	Column C
(table contents pending)		

Table repeatability: Mean and CV of SEA, peak force, and compaction efficiency across n replicate prints for the top- k designs.

5 Discussion

Populate Discussion section once Results are in. Anchor against the gaps identified in Section 1 and the prior work cited in Section 2.

Topics to address:

- Comparison to the simulation-driven multifidelity framework of Mo et al. [17] and the single-condition fabrication study of Pajunen et al. [11].

- Practical considerations: TPU–PLA interface durability under cyclic loading; sensitivity of optimal designs to print parameters (extrusion temperature, layer height); transferability to TPU/PETG or other high-temperature thermoplastics.

- Path forward toward integrated tread+lattice structures, miniaturization to crutch-tip envelopes (from the Edison crutch-tip synthesis), and field testing. Long-term crutch use is associated with substantial upper-extremity overuse injury and ground-reaction-force impulses [44–48]; existing spring-loaded [49, 50] and polymer-damped [51] tip designs reduce loading rate but not peak transmitted force, leaving an open design opportunity that an architected tensegrity-inspired insert is well-positioned to address. Recent open-source 3D-printed crutch work [52] and expert evaluations of orthopaedic caps [53] suggest a plausible translation path; adjacent footwear and orthotic lattice studies [54–56] provide additional transferable design heuristics.

Limitations..

6 Conclusions

We have presented an experiment-driven Bayesian-optimization framework for designing multi-material 3D-printed tensegrity-inspired energy absorbers using PLA struts and TPU tension elements. The framework parameterizes a family of unit cells, fabricates each candidate on a single multi-material FDM platform using a core-wrapping strategy [12], characterizes the response under quasi-static compression and drop-weight impact, and updates a Gaussian-process surrogate [21] that recommends the next batch of designs. The approach is broadly applicable to architected absorbers whose constitutive response is difficult to calibrate from first principles and where physical experiments—rather than simulations—are the natural ground truth.

Acknowledgment

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Funding Data

- Brigham Young University Mentored Research Grant ().
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Conflict of Interest

The authors declare no conflict of interest.

Pending TODOs and figure/table placeholders

- Tighten abstract to 150–200 words with concrete numerical results once the experimental campaign is complete. 1
- Manuscript status. At time of writing the project is at the *planned-methods* phase: the parameterized unit cell, multi-material print recipe, joint geometry, quasi-static and drop-impact protocols, and Bayesian-optimization loop described in Section 3 have been scaffolded and prototype hardware has been printed, but the closed-loop experimental campaign has not yet produced reportable results. Sections 4 and 5 therefore remain placeholders. 1
- Add 1–2 sentences clearly stating the gap this paper closes vs. Pajunen et al. [11] (single fabrication condition, no closed-loop optimization) and Mo et al. [17] (multifidelity BO, but simulation-only). 1

315	Decide between (a) co-Kriging or nonlinear information-
316	fusion priors [22] once pretensioned-assembly data are
317	available, or (b) keep this work single-fidelity and re-
318	port the pretensioned campaign separately.
319	Figure overview: Closed-loop, experiment-driven de-
320	sign framework: parameterized PLA/TPU tensegrity-
321	inspired unit cells are printed, tested under quasi-static
322	compression and drop-weight impact, and the resulting
323	performance data drive a Gaussian-process surrogate
324	that recommends the next batch of designs.
325	Cite Davami 2025 (<i>Addit. Manuf.</i>) and Intrigila 2022
326	(<i>Addit. Manuf.</i> 57:102946) once entries are added to
327	references.bib from PR #58 analysis.
328	Add a short paragraph on relevant TPU/PETG tensegrity
329	search-space bounds once the Edison literature task
330	5ae24eaf-5b6e-45cf-9f6c-1c7fbd881738 synthesis
331	is integrated.
332	Specify exact lower/upper bounds, manufacturing-
333	feasibility constraints, and the categorical-encoding
334	scheme used by the BO loop. Pull from the Edison
335	TPU/PETG search-space synthesis once available. . .
336	strut diameter
337	Cite the heterogeneous-parameters Edison LITERA-
338	TURE_HIGH brief (5191cf4d) on D3-symmetric vs.
339	fully-per-member parameterization.
340	Table design-vars: Design variables, units,
341	lower/upper bounds, manufacturing feasibility
342	constraints, and categorical encoding.
343	Document slicer profile (layer height, infill, print speed,
344	nozzle temperatures, retraction settings) and validation
345	prints used to characterize PLA-TPU interface strength.
346	Cite the joint-design Phase-3 CAD review (Edison
347	ANALYSIS 19e0c868) and the Phase-4 vision review
348	(e9a1f4cc) once both are integrated into the references.
349	Reference the manual-supports workflow (PR #66) and
350	the H2D multi-part assignment fix (PR #64) when
351	those branches land in main.
352	Figure fab-workflow: Fabrication workflow: param-
353	eter → CAD → slicer → multi-material print → post-
354	process and inspect.
355	Cite the Instron-stiffness Edison LITERATURE_HIGH
356	brief (9f68e71e) once the entry is added to refer-
357	ences.bib.
358	Cite the drop-test protocol document (PR #67) and
359	the drop-test Edison LITERATURE_HIGH brief
360	(653d7d39) once integrated.
361	Cite the M23 + Qtec equipment brief (Edison LITER-
362	ATURE_HIGH 1a0f4a70) once integrated.
363	Cite the per-modality objective-function briefs (Edi-
364	son LITERATURE_HIGH cfd30f3e, 7d6b43bf,
365	31126ee7, 9d74ab2e, f40e41a7) and the ANALYSIS
366	synthesis (789de8ab) once references.bib entries
367	are added from PR #60.
368	Specify the exact test fixtures, load-cell ranges, sampling
369	rate, DAQ, drop heights, and the protocol for averaging
370	replicate prints.
371	Add: kernel choice (e.g. Matérn-5/2), input warping,
372	normalization, batch size q , total budget T , initializa-
373	tion strategy (LHS or Sobol), and seed protocol. . .
374	Cite the BO scaffolding (PR #30), the T3-prism Sobol
375	batch (PR #35), and the heterogeneous-parameter Edi-
376	son brief (5191cf4d) once those branches land in main.
377	Figure bo-loop: Closed-loop BO schematic: design
378	proposal → multi-material print → compression and
379	drop-weight tests → GP update and acquisition maxi-
380	mization → next proposal.
381	Populate Results section once the experimental campaign
	produces data. Suggested subsections follow.

382	3	Plot best-so-far performance vs. number of physical ex-	383
		periments for each acquisition function, with confi-	384
		dence intervals over random seeds.	385
1	1	Figure convergence: Best-so-far SEA (and Pareto hy-	386
		per-volume for the multi-objective runs) vs. number	387
		of physical experiments, averaged over random seeds;	388
		baselines: LHS and random search.	389
		Show 3–5 representative geometries from along the	390
		Pareto front trading SEA against peak transmitted force.	391
1	1	Figure pareto: Pareto front in (peak transmitted force,	392
		SEA) space; representative geometries annotated. . . .	393
		Report between-print variability (CV) for top-ranked de-	394
2	2	signs; informs the choice of input-noise-robust acqui-	395
		sition.	396
		Table repeatability: Mean and CV of SEA, peak	397
2	2	force, and compaction efficiency across n replicate	398
		prints for the top- k designs.	399
		Populate Discussion section once Results are in. Anchor	400
		against the gaps identified in Section 1 and the prior	401
2	2	work cited in Section 2.	402
2	2	19–25 mm	403
		Identify limitations: design space size relative to exper-	404
		imental budget; reliance on a single fabrication plat-	405
		form; metric selection (axial impact only).	406
2	2	Two-sentence quantitative summary of the headline out-	407
		come (e.g., improvement in SEA over baselines within	408
		N rounds; identification of a Pareto-optimal geometry	409
		validated by replicate prints).	410
		Confirm exact grant numbers and acknowledge any	411
		NASA/Utah-Space-Grant-Consortium support per	412
3	3	acknowledgment-text guidelines.	413
		award number	414
3	3	Add Utah NASA Space Grant Consortium award if ap-	415
		plicable.	

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